



Analysis on the construction of sports match prediction model using neural network

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Abstract

To grasp the development of sports events in time and adjust the strategy in the process of match in time, the traditional back-propagation neural network (BPNN) algorithm was improved, and the match prediction model was constructed by using the adaptive BPNN. Taking the football match data of the Union of European Football Associations Champions League 2016–2017 as the prediction sample, the match was predicted. Moreover, taking some match data of Barcelona in 2016–2017 as an example, the fitting accuracy of the improved adaptive BPNN prediction model, multiple linear regression (MLR) model and grey degree prediction model were compared and analyzed. The research results showed that the prediction model built by the improved adaptive BPNN algorithm had smaller prediction error after rolling prediction. By comparing with the fitting accuracy of MLR model and grey prediction model, it was also found that the prediction error of the prediction model proposed was almost zero, and the error of the other two models was large. It showed that the prediction model proposed had high accuracy and reliability. Therefore, applying neural network to build a sports match prediction model and then predict the match results can provide a certain theoretical basis for sports match practice and the prediction and analysis of the match results.

Keywords BPNN · Prediction model · Sports match · Rolling prediction

1 Introduction

As the social economy develops and people's living standards improve, people focus more on material life and pursue more spiritual world. As the internet popularizes, people can get the results of all kinds of sports match in time. Whether it is live television or field watch, it will bring different experience to the audience. Among them, football is the largest single item in the world sports industry. It has a wide audience and great influence, making the annual GDP of the football industry in the world reach \$500 billion, which is known as the "17th largest economy in the world." While the proportion of football in the total output value of sports match is as high as 43%, far more than volleyball, basketball and other

items. Football leagues and clubs rank top in the global sports events and group business (Liu et al. 2018a).

According to statistics, the economic benefits of football industry are also very considerable. Whether from tourism or other aspects of consumption, football industry has made great contributions to the development of national economy (Mikołajczyk et al. 2018). There are many kinds of football leagues, including national competitions, such as Spanish Division One League, English Premier League and Chinese Super League, and continental competitions, such as Union of European Football Associations (UEFA) Champions League and Asian Champions League. Football's influence is also considered to be the first in the world, especially the four-year World Cup, which is also a sports event of global attention. It can not only promote the development of manufacturing industry, advertising industry, service industry and other industries, but also increase employment opportunities. Meanwhile, the tourism economy brought by football match is also very considerable. For China, it is estimated that by 2025, the development goals of China's sports industry and China's football industry will be 5 trillion yuan and 2 trillion yuan,

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respectively, which also shows the importance of football industry. Thus, the prediction of sports match also has some progress, including the economic prediction and the prediction of the sports match results. It can predict the development trend of the results of sports match scientifically and provide certain advice for strategic and tactical decision makers.

With the continuous development of science and technology, more and more machine learning technologies appear and help people to solve one problem after another. Back-propagation neural network (BPNN) is the most widely used form of neural network. It has the general advantages of neural network, including strong nonlinear mapping ability, high self-learning ability, generalization ability and error tolerance ability. When BPNN is used to predict in related fields, it also shows good performance and high prediction accuracy (McCann et al. 2018).

Therefore, in order to grasp the situation of the match in time and make strategic and tactical deployment in the process of the match, the traditional BPNN is improved, and a sports event prediction model is constructed. Then, the relevant data of the football match are used as a sample to predict, thereby providing more decision support for the football match and prediction models and ideas for other sports match. The BPNN model with powerful function of processing nonlinear data is selected. According to the demand, a three-layer structure BPNN prediction model is designed, including the design of the number of neurons in each layer, activation function, the type and number of input data, and how to predict the outcome of the game through the output, find its rules, and find a way to improve the level of winning and losing guessing from many sports games. On the basis of a large number of sports' data, this paper establishes a sports game prediction model and uses BPNN to train and predict the model. On the one hand, it enhances people's understanding of science and technology and sports; on the other hand, it effectively promotes sports knowledge and culture. The prediction results show that the BPNN model has high accuracy, which can provide a new modeling method for sports practice and theoretical basis for sports prediction analysis and research, and have a certain practical reference value for studying the development trend of various sports achievements in the future.

2 Literature review

With the development of computer technology, neural network algorithm shows good performance in many fields, especially in the prediction. Tang et al. (2018) proposed a lane changing predictor based on adaptive fuzzy neural network (FNN) to predict the steering angle. The results showed that this method could accurately track the steering

angle, and it also verified the better learning ability of the adaptive FNN method when compared with other machine learning methods. As for sensitivity analysis, it was also found that the travel angle and the vehicle acceleration were also the important factors in the prediction model (Tang et al. 2018). Liu et al. (2018b) used wavelet packet analysis, convolution neural network and convolution long-term and short-term memory network to propose a new wind speed prediction model, thus predicting the wind speed. The research results showed that the proposed model was robust and effective in predicting one-dimensional wind speed time series, and when the wind speed changed, the proposed model had better prediction performance than other models. Cortez et al. (2018) used recurrent neural network to predict the new architecture and then accurately predicted the emergency.

Thabtah et al. (2019) compared the performance of naive Bayes, artificial neural networks (ANNs) and decision trees with the models based on different feature sets of basketball, thus improving the performance of key features. The results showed that in the game results of the National Basketball Association (NBA), defensive fence was considered as the most important factor. Other factors were selected in the research process, such as three-point hit rate, free throw and total rebounds. The accuracy of the prediction results of the model proposed was increased by 2–4%, showing good performance (Thabtah et al. 2019). An et al. (2017) used the grey system theory to establish the GM (1, 1) prediction model and then predicted and analyzed the changes in the internal structure of the sports industry in 2016–2020. The results showed that the sports industry in China would have a good development in the next few years, and the prediction accuracy of the grey system theory was high, which well reflected the future development of the sports industry and provided certain basis for its development.

Souri et al. (2018) put forward a hypothesis that users with similar personalities will show mutual behavior patterns when they cooperate through social networks. Through the analysis of user activities in Facebook, they made personality recognition and collect information about users' personality characteristics and their personal data on Facebook. Based on all the data collected, different data mining technologies were used to learn the classifier, thereby identifying the user's personality through their personal data without filling in any questionnaire. By comparing the results of classifiers, the accuracy of the proposed decision tree model was 82.2%, which was more accurate than the previous research that could predict five factors of personality based on the variables in personal data, and these factors were used for the personality recognition model (Souri et al. 2018).

Souri and Hosseini (2018) outlined the current challenges related to malware detection methods in data mining, systematically and categorically summarized the current machine learning mechanism methods and discussed the important factors of malware classification methods in data mining. According to the important factors of detection methods, they were compared. The advantages and disadvantages of data mining model, evaluation method and proficiency were discussed. This survey helped researchers fully understand the field of malware detection and allowed experts to carry out follow-up inspection (Souri and Hosseini 2018). Hu et al. (2019) used ANN and data analysis to enable NBA team to improve training quality, enrich tactics, improve team performance and enhance brand influence. Based on the ANN, the most valuable player in NBA was modeled and predicted, which provided reference for improving and promoting the research of using ANN and data analysis to evaluate athletes in China.

To sum up, the previous studies used neural network algorithm to predict some diseases, output or stock price. However, for sports, other algorithms are applied to find out the influencing factors of the competition or predict the future development of the industry, but rarely use the neural network algorithm to predict the match results. Therefore, BPNN is used here to build the prediction model of sports match and then predict the match situation, which is helpful to adjust the tactics in time and win.

3 Method

3.1 BPNN

In the mid-1980s, the emergence of BPNN solved many problems. It was also proved that BPNN had the ability of arbitrary and complex pattern classification and good mapping of multiple functions. It is one of the most widely used neural network models with the best performance. It is a multilayer feedforward neural network trained by error BP algorithm. At present, in practical application, BPNN is mainly used in function approximation, pattern recognition, classification and data compression (Armenteros et al. 2019; He and Zhang 2018; Kumar and Singh 2018).

BPNN uses gradient search technology to minimize the error mean square deviation between the actual output value and the expected one. The basic BPNN algorithm includes two processes: the forward propagation of signal and BP of error. BPNN usually consists of three important parts: input layer, hidden layer and output layer. The learning of BPNN is a supervised learning process. The model constantly adjusts the parameters in the neural network by learning BP error to correct the errors and finally

realizes or approaches the expected mapping relationship between input and output. The learning of BPNN is divided into two stages (Wang and Li 2018; Muralitharan et al. 2018). Figure 1 shows the structure diagram of BPNN.

First is the stage of forward propagation. The input signal enters the input layer, reaches the output layer after learning from the hidden layer and then outputs the results. In the process of information forward propagation, the weight of neurons remains unchanged, and the output of each layer of neurons only affects the input of the next layer of neurons. When the error between the final output information and the supervision signal is within the preset threshold, it will continue completing the learning process; otherwise, it will carry out BP. Second is the BP stage. In this stage, the error signal in the network is transmitted from the output layer to the input layer. In the process of error signal transmission, the weights between the neurons in the network will have error signals for feedback adjustment, and the weights will gradually stabilize in the adjustment process. Finally, BPNN will gradually stabilize to get the most ideal output. The forward propagation and BP are repeated and will not stop until the error is less than the set value (Yu et al. 2018; Chen 2019; He and Zhang 2018).

In the input layer of BPNN, the input variable is x , where $x = (x_0, x_1, \dots, x_{n-1})$. The hidden layer has n neurons whose output is $x = (x_0, x_1, \dots, x_{n-1})^T$; there are m neurons in the output layer and $y = (y_0, \dots, y_{m-1})^T$. The weight from input layer to hidden layer is W_{ij} , the threshold value is θ_j , while the weight from hidden layer to output layer is w_{jk} , and the threshold value is θ_k (Zhang et al. 2018; Zhang et al. 2019; Liang et al. 2018). Therefore, the output neurons are obtained (Eq. 1):

$$\begin{cases} x_j = f\left(\sum_{i=0}^{n-1} w_{ij}x_i - \theta_j\right), & j = 0, 1, \dots, n-1 \\ y_k = f\left(\sum_{i=0}^{n-1} w_{ik}x_i - \theta_k\right), & k = 0, 1, \dots, m-1 \end{cases} \quad (1)$$

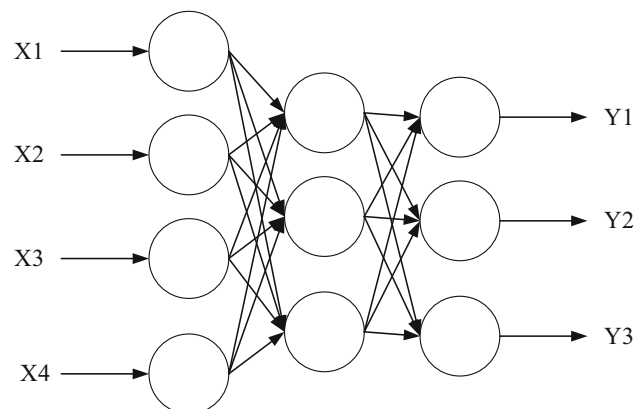


Fig. 1 The structure of BPNN

The activation function $f(x)$ is a sigmoid function (Eq. 2):

$$f(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

3.2 Improved BPNN algorithm

The traditional BPNN algorithm adopts the gradient steepest descent method, which usually takes the error square as the objective function. Therefore, it may lead to the situation that the training falls into the local minimum value and slows the convergence speed of learning, making it unable to guarantee the number of hidden layers and nodes and thus affecting the generalization ability of BPNN algorithm (Hughes et al. 2018).

Hence, the traditional BPNN is improved and the adaptive variable step size algorithm is used to improve the convergence speed of the algorithm (Colston et al. 2018; Bruce et al. 2018). In the traditional BPNN, the fixed step size is λ , so the improved BPNN is obtained by using adaptive variable step size (Eq. 3):

$$W(k+1) = W(k) + \alpha D(k) \quad (3)$$

$W(k)$ represents the weight at k , and α is learning rate, while the negative gradient at k can be expressed by Eq. 4:

$$D(k) = -\frac{\partial E}{\partial W(k)} \quad (4)$$

The learning rate at k is shown in Eq. 5, and the step length is shown in Eq. 6:

$$\alpha(k) = 2^{\lambda} \alpha(k-1) \quad (5)$$

$$\lambda = \text{sign}[D(k)D(k-1)] \quad (6)$$

Therefore, Eq. 3 is adjusted to Eq. 7:

$$W(k+1) = W(k) + \alpha(k)[1 - \eta D(k) + \eta D(k-1)] \quad (7)$$

η is the momentum factor, used to suppress the oscillation.

The adaptive adjustment of learning rate can effectively shorten the learning time, and there is an equation for adjusting learning rate in MATLAB running software (Eq. 8).

$$\alpha(k+1) = \begin{cases} 1.05\alpha(k), & \text{mse}(k+1) < \text{mse}(k) \\ 0.7\alpha(k), & \text{mse}(k+1) > 1.04\text{mse}(k) \\ \alpha(k), & \text{others} \end{cases} \quad (8)$$

where mse is mean square error. Once the training reaches the maximum training times or the error meets the requirements, BPNN stops learning.

3.3 Influencing factors of match results

In the match, there are many factors that affect the result. According to the previous research of scholars, the relationship between these factors that affect the result is complex and cannot be expressed by equations. In summary, there are several factors as follows, goals, goals lost, the rate of ball control, home court advantage and shots (Carey et al. 2018).

Among them, goals are closely related to the result of the game, while shots, corners and goals are directly related to each other, showing a positive correlation (Zaveri et al. 2018). However, as another major factor affecting the outcome, goals are not closely related to shots, positive shots (rate) and corner shots. It is generally considered to be related to steals and offside traps, and it is a negative correlation. In addition, taking Barcelona as an example, it has a strong ability to control the ball, making it always ranked in the forefront of the world's football clubs (Ujwal et al. 2018). The match between the teams in the Chinese Super League shows that the weaker teams can also compete with the strong opponents by means of the passing control technology. Therefore, the ball control rate is also considered to be indirectly related to goals and goals lost (Chalmers et al. 2018).

In addition to the players' own technology, home court advantage has also become a factor affecting the result of the game. This is because when playing, the home team is very familiar with the environment, and there are many fans who can cheer for the team members; while the visiting team will reduce the combat effectiveness because of the tiredness of the journey, or even the acclimatization. Meantime, there will be other factors such as the referee's preference for the home team. These factors interact with each other, forming the home court advantage, which, to a certain extent, also affects the match results. Data showed that in the 2016–2017 season, in the 30 rounds of 240 games in the Chinese Super League, the home team won 189 games and lost 51 games, which was lower than 25% of the total games. This is also enough to show that home court advantage has a certain impact on the results of the game (Constantinou 2019).

In the process of the match, in addition to goals, goals lost, the rate of ball control and home court advantage, there are also some match data that will affect the results, such as the positive correlation between shots and goals (Chen 2019). The number of shots directly affects positive shots, and the number of corners and passes also has a certain correlation with shots. Different factors have a positive or negative correlation or an indirect relationship with the outcome of a competition, so when predicting the

match results, multiple eigenvalues need to be considered to improve the prediction accuracy (Cho et al. 2018).

Of course, in addition to the factors in the match, the environment will also affect the results, such as the weather conditions on that day, the state of the players, the cooperation between the players and the coach and the strategy and tactics of the coach. Some cannot be known in advance, which to a certain extent will also affect the match results (Shen et al. 2019; Chen and Shen 2019).

3.4 Construction of prediction model and parameter setting

To ensure the influence of the hidden layer on the BPNN, a prediction model is built by using BPNN and a three-layer structure is adopted. Therefore, in parameter design, the number of neurons in input layer, hidden layer and output layer should be determined first. In 2016–2017 season, the number of goals, shots, positive shots, passes, corners, number of crosses, steals, ball control rate, home and away situation, player's competitive state and other variables are taken as variables, that is, the number of neurons in the output layer (Liu and Wang 2019). As neurons in the hidden layer will directly affect the performance of the whole network, it is necessary to determine the number of hidden layer neurons based on Eqs. 9–11 to avoid the occurrence of transition fitting and to reduce neurons in the hidden layer on the premise of satisfying the prediction accuracy.

$$m = \sqrt{n + l} + \alpha \quad (9)$$

$$m = \log_2 n \quad (10)$$

$$m = \sqrt{nl} \quad (11)$$

m , n and l represent the number of neurons in the hidden layer, input layer and output layer, respectively, α represents a constant between [1, 10], and l needs to be determined according to the prediction. In the match result prediction model, it is supposed to predict the probability of a team winning, and the probability value is between 0 and 1. Therefore, the number of neurons in the output layer should be 1 at last.

The activation function can deal with different data in different ways, and it will also affect the prediction model. Hence, it is necessary to select different activation functions in BPNN to obtain more appropriate data. The input data in the input layer are all discrete data, which do not need to be transformed by function, and at this time, the activation function is linear function. The data of the hidden layer need to deal with its nonlinear relationship. In other words, using the Sigmoid function mentioned above can make the signal gain effect expression of the middle position, and the signal gain of both ends is small, thereby

ensuring the mapping effect. Moreover, the input preferences can be weighted to generate nonlinear decision boundaries for guaranteeing that the final mapping results are ranged [0, 1], and that the data are not easy to diverge in the conversion process. In the final output layer, because the result is a probability value, the Tanh activation function is needed to ensure that the final variable can be mapped to the interval of [0, 1], and then the activation function at this time is shown in Eq. 12:

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (12)$$

When the input value of Sigmoid function is valued $[-1, 1]$, the function value obtained is sensitive to change; when the output value is close to or beyond the range, it will lose its sensitivity, thus being in a saturated state and affecting the accuracy of neural network prediction. Tanh's output and input can keep a nonlinear monotonic ascending and descending relationship, which is in line with the gradient solution of BPNN, has a good fault tolerance between 0 and 1 and conforms to the law of human brain neural saturation.

Therefore, there are 19 neurons in the input layer and 1 neuron in the output layer. The initial values are all 0, 0.2 for a win, 0.1 for a draw and 0 for a negative. Take momentum factor $\eta = 0.3$, initial learning rate $\alpha(1) = 0.01$, randomly assign the threshold value of the improved BPNN, and take the control error 0.0001.

Let the home team's and the visiting team's victory indexes be x and y , respectively, and the final winning and losing criteria are shown in Table 1.

In the design of BPNN model, the following factors are considered: the number of layers, the number of neurons in each layer, the selection of initial value, learning rate and expected error. A three-layer BP network can map from n -dimension to m -dimension arbitrarily, and the accuracy of the network can be improved by increasing the number of neurons in the hidden layer. The number of neurons in input layer and output layer depends on the complexity of solving specific problems. To improve the accuracy of network training, a hidden layer can be used to increase the number of hidden layer neurons. This design is simpler than increasing the number of hidden layer neurons. However, there is no definite equation to determine the

Table 1 Criteria for judging the outcome of an event

Index value	Prediction results
$x > y$ and $1 > x > 0.5$	Home team wins
$y > x$ and $1 > y > 0.5$	Visiting team wins
$x < 0.5$ and $y < 0.5$	It ends in a draw

number of hidden layer neurons. As the volatility of the forced function increases, the number of hidden layer neurons should be increased appropriately. In the selection of the initial value, the output value of each neuron weighted by the initial value is generally close to zero, so that the connection weight of each neuron can be adjusted at the place where their S-type activation function changes the most. Hence, the initial value is generally a random number between $(-1, 1)$. For any network, there is a suitable learning rate. The learning rate determines the amount of weight change in each cycle training. The large learning rate can lead to the instability of the network, but the small learning rate can lead to the extension of the training time and the slow convergence speed, which cannot guarantee that the error of the network can finally be minimized. To ensure that the data are the same order of magnitude, it is necessary to preprocess the input and output data of the neural network to speed up the network training. Figure 2 shows the prediction model constructed by the improved BPNN.

Rolling prediction is based on single-step prediction. Single-step prediction refers to that when $k = 1$, the network input x value has m historical data, and finally output

x_{n+m+1} . Further predict the value of the future k time and input x value through k iterations. Each time rolling to the next value, it is capable of outputting different prediction results.

4 Results and discussion

4.1 Data collection and processing

Taking the data of the two teams in the **2016–2017 UEFA Champions League as samples**, the improved BPNN prediction model is predicted, and the data are shown in Table 2.

In the sample data, the victory probability of home court in the fourth and sixth matches is 1, while that in the fifth, seventh and eighth matches is 0.5. Due to the difference of data digits, it is necessary to convert these digits to $[0, 1]$ for normalization processing. The conversion function is shown in Eq. 13 to facilitate prediction.

$$y = a \tan(x)^{*} 2 / \pi \quad (13)$$

The data in Table 3 are obtained.

Fig. 2 Modeling process of BPNN prediction model

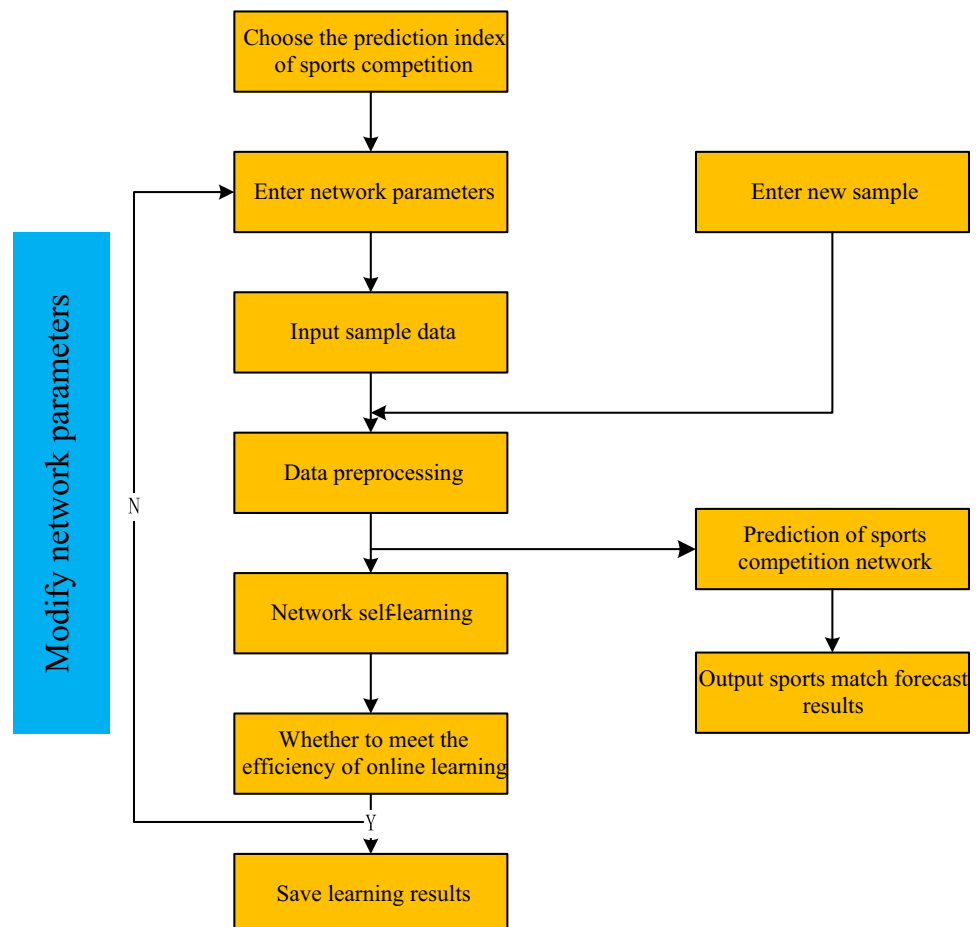


Table 2 Sample data

Scene	4	5	6	7	8
Home court	1.0	0.0	1.0	0.0	1.0
Home team's goals	1.70	1.9	1.5	1.5	1.5
Home team's shots	11.70	12.25	13.0	12.50	12.50
Home team's positive shots	5.0	4.75	4.0	3.75	5.25
Home team's passes	447.25	441.75	461.25	435.5	404.5
Home team's offside raps	0.75	1.0	1.5	1.7	2.0
Home team's corner count	5.0	3.75	4.5	4.0	4.0
Home team's steals	12.75	13.0	12.75	13.75	11.6
Home team's ball possession	49.3	52.25	55.25	55.25	53.6
Status of home team's member	0.7	0.8	0.9	0.7	0.7
Visiting team's goals	1.65	0.35	1.1	0.6	1.0
Visiting team's shots	16.75	11.25	8.50	12.90	10.50
Visiting team's positive shots	7.10	3.75	3.5	6.75	5.5
Visiting team's passes	367.25	450.75	310.25	400.5	319.5
Visiting team's offside raps	2.75	1.5	1.8	1.8	2.4
Visiting team's corner count	5.25	4.75	3.5	4.0	5.0
Visiting team's steals	15.75	18.0	14.75	11.75	15.6
Visiting team's ball possession	51.3	52.5	40.25	55.5	40.6
Status of home team's member	0.5	0.2	0.3	0.4	0.5
Victory or defeat	1.0	0.5	1.0	0.5	0.5

Table 3 Data converted

Scene	4	5	6	7	8
Home court	1.000	0.000	1.000	0.000	1.000
Home team's goals	0.7095	0.6675	0.5979	0.5979	0.5979
Home team's shots	0.9525	0.9484	0.9596	0.9600	0.9482
Home team's positive shots	0.8709	0.8577	0.8526	0.8705	0.8696
Home team's passes	0.9991	0.9975	0.9925	0.9925	0.9875
Home team's offside raps	0.4405	0.5000	0.5979	0.7152	0.7038
Home team's corner count	0.9107	0.8175	0.8577	0.8760	0.8760
Home team's steals	0.9175	0.9409	0.9705	0.9752	0.9655
Home team's ball possession	0.9831	0.9825	0.9925	0.9825	0.9961
Status of home team's member	0.7000	0.8000	0.9000	0.7000	0.7000
Visiting team's goals	0.5565	0.2235	0.5571	0.4301	0.5000
Visiting team's shots	0.9675	0.9625	0.9250	0.9606	0.9450
Visiting team's positive shots	0.8910	0.8875	0.8501	0.9075	0.9125
Visiting team's passes	367.25	450.75	310.25	400.5	319.5
Visiting team's offside raps	0.7525	0.5597	0.6808	0.8131	0.500
Visiting team's corner count	0.8825	0.8775	0.7579	0.8755	0.8944
Visiting team's steals	0.9675	0.9720	0.9875	0.9475	0.9596
Visiting team's ball possession	0.9973	0.9895	0.9825	0.9885	0.9876
Status of home team's member	0.5000	0.2000	0.3000	0.4000	0.5000
Victory or defeat	1.0000	0.5000	1.0000	0.5000	0.5000

The MATLAB software is used to analyze the converted data. When the number of learning times is 1000, the winning situation of the home team is predicted, and the winning rate of the home team is about 88%.

4.2 Discussion

By running MATLAB software, and using Eq. 9, the value range of m obtained is [5, 15], and by using Eqs. 10 and 11,

m is 4. Therefore, set different number of hidden layers and then find the most appropriate m value.

When m is 5 and the learning rate is 0.01, the maximum number of training steps is 40, and the training situation is shown in Fig. 3.

When m is 10 and 15, the learning rate is 0.01, and the maximum number of training steps is 40. Figure 4 shows the training situation.

When analyzing the number of neurons in different hidden layers of Figs. 3 and 4, it is found that when the number of neurons in hidden layer is 5, the speed of network training is the fastest and the performance is the best.

According to the data, in the first round of the 2016–2017 season's top 16 matches, Manchester City, as the host team, competed with the visiting team Monaco and finally got a score of 5:3. Therefore, the error of the prediction model is analyzed and Fig. 5 is obtained.

Figure 5 suggests that the error of the prediction model is almost 0, indicating that the training accuracy of the sample data is high and the data fitting is also more accurate.

Some characteristic data of Fútbol Club Barcelona in the 2016–2017 season are tested by using the prediction model, and the data are shown in Table 4.

The prediction accuracy of BPNN, multiple linear regression (MLR) model and grey prediction model are compared and analyzed by using the data, and the accuracy results in Table 5 are obtained.

Table 5 shows that the prediction result error of the improved adaptive BPNN is the smallest, and the accuracy of the improved adaptive BPNN prediction model is the highest, so the improved adaptive BPNN prediction model is reliable.

To improve the accuracy and effectiveness of the prediction, five rounds of competition are predicted through

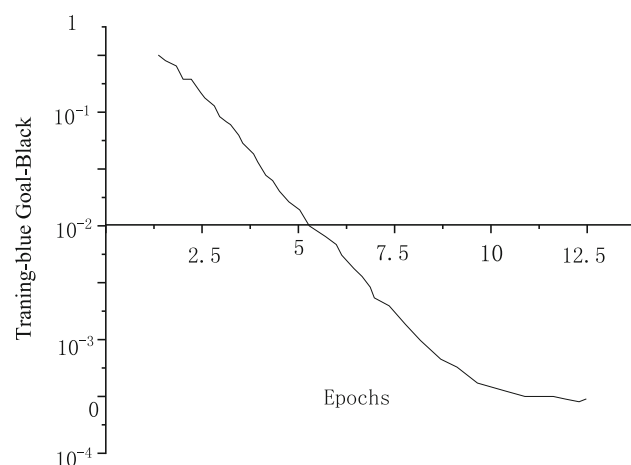


Fig. 3 The training results when the number of neurons in the hidden layer is 5

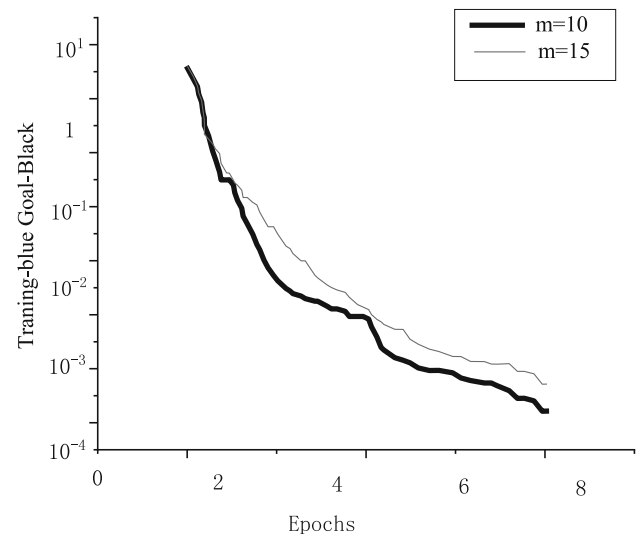


Fig. 4 Training results of neurons in different hidden layers

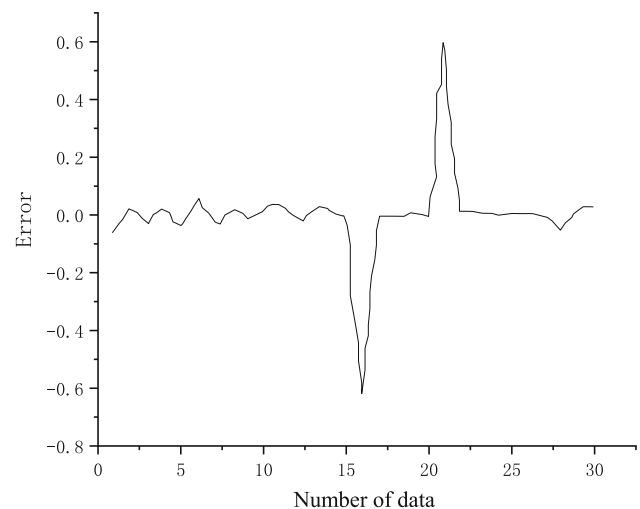


Fig. 5 Prediction error curve

several tests. The average accuracy of BPNN to predict the outcome of a football match is 66.5% or more. As shown in Table 6, according to the data comparison and analysis with the actual situation, some possible reasons for the inconsistency or the opposite are summarized: the training data set dimension is too small, resulting in the decline of training accuracy; there are various uncertain factors in the actual competition, such as weather and suspension of players, which may lead to the inconsistency between the results and the expected results.

The BPNN of the sports match prediction system is designed, and the factors that affect the outcome of the game and their relationship are analyzed. The results show that the BPNN is practical to the sports match prediction problem and has strong ability to solve the nonlinear relationship problem. The fitting accuracy of the prediction

Table 4 Some features of Barcelona competition

Goals for	Corner count	Positive shots	Free throw	Victory or defeat
1	4	4	8	Win
2	2	5	5	Win
1	15	8	8	Draw
1	6	11	2	Draw
1	2	6	12	Lose
1	4	4	6	Lose

Table 5 Fitting accuracy of prediction model

Improved adaptive BPNN		MLR		Grey prediction model	
Predicted value	Error	Predicted value	Error	Predicted value	Error
2 wins and 3 losses	− 0.001	2 wins and 1 loss	0.1001	2 wins and 0 loss	0.1102

Table 6 Comparison between actual results and predicted results

Round	Actual results	Predicted results	Accuracy (%)
6	Draw, lose, draw, draw, win, lose, win, lose	Win, lose, lose, draw, draw, lose, win, lose	57.5
7	Lose, win, win, win, lose, win, draw, win	Win, win, win, win, lose, win, draw, raw	75
8	Win, draw, draw, win, draw, lose, lose, lose	Lose, draw, draw, draw, draw, lose, win, lose	75
9	Win, win, draw, draw, draw, draw, draw, win	Lose, win, draw, win, draw, draw, draw, win	75
10	Lose, draw, win, win, lose, win, lose, draw	Lose, lose, win, win, draw, win, draw, win	50

model is analyzed from the characteristic data of sports competition in the prediction model. BP network has strong nonlinear mapping ability and flexible network structure, and the number of processing units and network learning coefficient of each layer in the middle layer of the network can be set arbitrarily according to the specific situation and get different performance. Whereas, in essence, BP algorithm is a gradient descent method, the performance of the algorithm depends on the initial conditions, and the learning process is easy to fall into local minima. Numerical simulation results show that the learning speed, accuracy, initial robustness and network generalization performance of BP algorithm are poor, which cannot meet the application requirements.

Table 7 gives the error index and certainty coefficient of three single models and the BPNN combination model established. According to Table 6, for MSE, MAE and

MAPE with smaller and better index, the prediction effect of four models from excellent to inferior is BPNN combination model, BPNN model, MLR model and DCCA, respectively; for DC with larger and better index, the prediction effect of four models from superior to inferior is BPNN combination model, BPNN model, MLR model and DCCA, respectively. To sum up, judging from the evaluation index of model validation period, the BPNN combination model based on MLR-DCCA has better prediction effect than single model.

In BPNN model, as shown in Table 8, RMSE in training and prediction stage is 0.22 and 0.34, respectively, MAPE is 7.60% and 9.21%, respectively, and MSE is 0.89 and 0.85, respectively. The accuracy and stability of this model are much higher than that of BPNN model and stepwise regression model. Meanwhile, the RMSE and MAPE indexes show that the accuracy of the model is the highest

Table 7 Evaluation index value of model validation period

Model category	Evaluation index			
	MSE	MAE	MAPE	DC
MLR model	13.948	16.901	0.122	0.789
BPNN model	4.461	5.630	0.086	0.989
De-trend cross-correlation analysis (DCCA)	15.525	21.270	0.184	0.661
BPNN combination model	3.342	4.067	0.064	0.994

Table 8 Analysis and comparison of simulation and prediction of performance parameters

Model	RMSE			MAPE				MSE				
	Prediction	Training	Maximum	Minimum	Prediction (%)	Training (%)	Maximum (%)	Minimum (%)	Prediction	Training	Maximum	Minimum
BPNN prediction model	0.34	0.22	0.32	0.11	9.21	7.60	8.59	9.59	0.85	0.89	0.61	0.44
	0.40	0.27	0.35	0.40	13.42	10.47	12.75	19.58	0.84	0.79	0.50	0.34
	0.48	0.35	0.49	0.51	21.48	13.50	10.58	32.66	0.69	0.74	0.93	0.94

among the three models, and the MSE is the largest, which further proves that the stability of the model is the best. The reason is that the BPNN model cannot adjust the weight and threshold of the hidden layer in the process of prediction and fitting, and it will fall into some problems such as local minimum, slow convergence speed and over-fitting. For this reason, the hidden layer of BPNN is optimized. Additionally, BP model gives full play to the generalization mapping ability of BPNN, further improves the prediction ability and stability and fully verifies the above results from the evolved weight coefficient and network structure through the optimization of training samples.

To sum up, the error of the prediction results is almost zero, whether it is based on multiple factors influencing the results of the UEFA Champions League or some features of Barcelona in 2016–2017 season. Compared with the prediction results of MLR model and grey prediction model, the improved adaptive BPNN has higher fitting accuracy and better performance.

5 Conclusion

Based on the improvement in the traditional BPNN, an adaptive BPNN algorithm prediction model is constructed. Taking the football match data of the 2016–2017 UEFA Champions League as the sample, the validity of the prediction model is verified. Moreover, the improved adaptive BPNN is compared with MLR and the grey prediction model in terms of the fitting accuracy through the match characteristic data of Barcelona in 2016–2017. Furthermore, the validity and reliability of the improved adaptive BPNN model are proved. However, the noise of the sample data is not processed, which may have a certain impact on the prediction results. Therefore, the weighted voting method will be used to reduce the impact of the noise in the later research. Using neural network algorithm to build sports match prediction model can predict the results more accurately, provide certain theoretical basis for sports match practice and the prediction and analysis of sports match results and provide basis for formulating reasonable and long-term goals. But there are also many deficiencies in practice that the function is not perfect, and the final accuracy is not high enough. To improve the existing prediction model as much as possible, it is necessary to learn the algorithm of initial weight optimization of BPNN in the future, improve the accuracy of prediction and strengthen the ability of demand analysis and system architecture design in software engineering.

Compliance with ethical standards

Conflict of interest All authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors. Informed consent was obtained from all individual participants included in the study.

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